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# Bottleneck Games in Noncooperative Networks

Ron Banner and Ariel Orda
Department of Electrical Engineering
Technion – Israel Institute of Technology
Haifa 32000, Israel
{banner@tx, ariel@ee}.technion.ac.il

*Abstract*—**We consider routing games where the performance of each user is dictated by the worst (bottleneck) element it employs. We are given a network, finitely many (selfish) users, each associated with a positive flow demand, and a load-dependent performance function for each network element. We first prove the existence of a Nash equilibrium, considering two routing scenarios, namely when a user can split its traffic over more than one path (splittable bottleneck game) and when it cannot (unsplittable bottleneck game); we also consider the convergence properties of each game. Then, we turn to investigate the efficiency of the Nash equilibria in both games with respect to the network optimum in terms of bottleneck performance; specifically, while for both games we show that the *price of anarchy* is unbounded, we identify for each game conditions under which Nash equilibria are optimal. Finally, we analyze for each game the performance deterioration at Nash equilibria with respect to the *additive* network performance objective of "total cost".**

***Keywords*- bottleneck & additive metrics, Nash equilibrium, price of anarchy, price of stability, selfish routing, unregulated traffic.**

## I. INTRODUCTION

Traditional computer networks were designed and operated with systemwide optimization in mind. Accordingly, the actions of the network users were determined so as to optimize the overall network performance. Consequently, users would often find themselves sacrificing some of their own performance for the sake of the entire network. However, it has been recognized that systemwide optimization may be an impractical paradigm for the control of modern networking configurations [1],[13],[17],[22],[23],[26]. Indeed, control decisions in large scale networks are often made by each user independently, according to its own individual interests. Such networks are henceforth called *noncooperative*, and Game Theory [18] provides the systematic framework to study and understand their behavior.

Game theoretic models have been employed in various contexts, such as flow control [3],[13],[26], routing [17],[22],[23] and bandwidth allocation [15]. These studies mainly investigated the structure of the network operating points i.e., the Nash equilibria of the respective games. Such equilibria are inherently inefficient [11] and, in general, exhibit suboptimal network performance. As a result, the question of how much worse the quality of a Nash equilibrium is with respect to a centrally enforced optimum has received considerably attention e.g., [1],[10],[14],[16],[21],[22],[23]. In order to quantify this inefficiency, two conceptual measures have been proposed in the literature. The first, termed the *price of anarchy* [19], corresponds to a worst-case analysis and it is the ratio between the *worst* Nash equilibrium and the social optimum. The second, termed the *price of stability* [2] (or the *optimistic price of anarchy* [1]) is the ratio between the *best* Nash equilibrium and the optimum, and it quantifies the degradation in performance when the solution is required to be *stable* (i.e., with no agent having an incentive to independently defect out of it once being there).

The above studies focused on the case where the structure of the user performance objective is *additive* i.e., performance is determined by the *sum* of link cost functions. Yet, another fundamental case is that of *bottleneck objectives* (also known as *Max-Min* or *Min-Max* objectives), in which performance is determined by the *worst* component (link). Accordingly, in this study we investigate the case where users route traffic selfishly so as to optimize the performance of *their* bottleneck elements, given the routing strategies of all other users. Such settings give rise to a non-cooperative game, which is henceforth termed *the bottleneck game*.

Bottleneck games emerge in many practical scenarios. For example, they are encountered in wireless networks, where each node has a limited battery (i.e., transmission energy) so that the node's lifetime depends on the total flow that emanates from it. In this setting, each user would route its flow demand so as to maximize *its* connection's lifetime (i.e., the time until the *first* battery along its routing topology drains out) while the social objective is to maximize the *network lifetime* (defined as the time until the *first* battery in the *network* drains out [9]). Bottleneck games also arise when users attempt to enhance the ability to accommodate momentary traffic bursts by maximizing the residual capacities of the links they employ. In these cases, the network typically aims at maximizing the *common scaling factor* [5] (i.e., a common factor that can scale up the traffic demands of all users without violating the capacity constraints), whereas each user aims at maximizing the factor that its own traffic demand can be scaled with. Traffic engineering is another major framework where bottleneck objectives (and bottleneck games) are encountered. For example, in view of the limited sizes of transmission buffers at nodes, it is desirable to minimize the utilization of the most utilized buffers in order to avoid deadlock and reduce buffer space [6]. Similarly, in congested networks it is desirable to minimize the utilization of the most utilized links so as to move traffic away from congested hot spots to less utilized parts of the network [4],[25].

As mentioned, in spite of their fundamental importance, bottleneck games have not been considered in the literature. Accordingly, in this study we consider two classes of bottleneck games. In the first, each user can split its traffic among any number of paths (henceforth, *splittable bottleneck game*), while in the second, each user routes its traffic along a single path (henceforth, *unsplittable bottleneck game*).

## Our Results

First, we prove that, for both splittable and unsplittable bottleneck games, there is a (not necessarily unique) Nash equilibrium; we note that, for the splittable case, a major complication is the inherent discontinuity of the objective functions. Then, we turn to consider the rate of convergence of *best response dynamics* in both games. For unsplittable bottleneck games, we show that convergence to a Nash equilibrium is always achieved within a finite number of steps; moreover, when the number of users is small (i.e., constant), convergence time is polynomial in the network size. On the other hand, for splittable bottleneck games, we show that the convergence time may be unbounded.

Next, we investigate the efficiency of the Nash equilibrium. Specifically, for each game we compare the network bottleneck (i.e., the performance of the worst element in the network) at Nash equilibrium with the optimal value (in terms of bottleneck objectives). In particular, for both games, we analyze the price of anarchy and identify conditions under which Nash equilibria are efficient.

More specifically, for the unsplittable case, we show that the price of anarchy is unbounded for the general case and is $O\left(M^{p}\right)$ in $M$-link networks with link performance functions that are polynomial with a degree $p$; we also give an example showing that the latter result is tight. Yet, we show that a *best* Nash equilibrium coincides with the *social optimum*; hence, *the price of stability is 1*. Unfortunately, we show that calculating a best Nash equilibrium is NP-hard.

Consider now the efficiency of Nash equilibria in splittable bottleneck games. Here too, we show that the price of anarchy is unbounded. Yet, we show that, if at a Nash equilibrium the users route their traffic along paths with a minimum number of bottleneck links, the Nash equilibrium is socially optimal. This finding might motivate the employment of some pricing mechanism that penalizes the use of paths with an excessive number of bottleneck links.

Finally, we consider the case in which the social (i.e., network) objective is additive, namely, minimize the *sum* of all link performance functions (the *total cost*), while the (selfish) users still consider *bottleneck* objectives. This is the case when the network is interested in minimizing the *average* link performance rather than the performance of the worst link. Here too, we show that the price of anarchy[1] is unbounded in both games. However, we show that the price of anarchy is upper bounded by the number

[1] Which, in this context, is the worst-case ratio between the *total cost* of a flow at Nash equilibrium and the minimum value of the *total cost*.

of network links when users splittably route their traffic along paths with a minimum number of bottlenecks. Again, we give an example that shows that this result is tight.

### Organization

The rest of this paper is organized as follows. In section 2, we formulate the model and terminology. In section 3, we establish the existence (and non-uniqueness) of splittable and unsplittable bottleneck games. In section 4, we consider the convergence properties of both games. In section 5, we investigate the efficiency of Nash equilibria for each of the two games. Finally, in section 6, we investigate the performance loss with respect to the additive network objective of total cost.

## II. MODEL AND TERMINOLOGY

We consider a finite set of users *U*, which share a communication network that is modeled by a graph $G(V,E)$. Each user $u \in U$ is associated with a positive throughput demand $\gamma_u$ and a pair of source-destination nodes $(s_u, t_u)$. We denote by $P^{(s_u,t_u)}$ the set of all paths from the source $s_u$ to the destination $t_u$ and by *P* the set of all the paths in the network; assume that the source and destination nodes of each user are connected i.e., $\left|P^{(s_u,t_u)}\right| \geq 1$ for each $u \in U$ .

A user $u \in U$ needs to send $\gamma_u$ units of flow from $s_u$ to $t_u$ along the paths $P^{(s_u,t_u)}$. We denote by $f_p^u$ the flow of user $u \in U$ on a path $p \in P^{(s_u,t_u)}$. A user *u* can assign any value to $f_p^u$, as long as $f_p^u \geq 0$ (non-negativty constraint) and $\sum_{p \in P^{(s_u,t_u)}} f_p^u = \gamma_u$ (demand constraint); this assignment of traffic to paths shall also be referred to as the *user strategy*. The set of all possible strategies of a user is referred to as the *user strategy space*. The product of all user strategy spaces is termed the *joint strategy space*; each element in the joint strategy space is termed a (flow vector) *profile*; effectively it is a global assignment of traffic to paths that satisfies the demands of all users.

Given a profile $f = \{f_p^u\}$ and a path $p \in P$, denote by $f_p$ the total flow that is carried over *p* i.e., $f_p = \sum_{u \in U} f_p^u$; also, denote by $f_e^u$ the total flow that user *u* transfers through *e* i.e., $f_e^u = \sum_{p|e \in p} f_p^u$ . Finally, for a flow vector *f* and a link $e \in E$, denote the total flow carried by *e* as $f_e$ .

We associate with each link $e \in E$ a *performance function* $q_e(\cdot)$ that depends on the total flow $f_e$ carried over *e*. We assume that, for all $e \in E$, $q_e(f_e)$ is continuous and increasing in $f_e$ . We define the *network bottleneck* $B(f)$ of a flow *f* as the performance of the worst link in the network i.e., $B(f) \triangleq \max_{e \in E} \{q_e(f_e)\}$. Similarly, we define the *bottleneck of a user* $u \in U$ as the performance of the worst link that *u* employs i.e., $b_u(f) \triangleq \max_{e \in E | f_e^u > 0} \{q_e(f_e)\}$.

Users are selfish i.e., each minimizes its *own* bottleneck. Since the bottleneck of a user $u \in U$ depends on the flow configuration of *all* users, we are faced with a *non-cooperative game* [18]. We consider two types of games. The triple $\langle G, U, \{q_e\} \rangle$ is termed a *splittable bottleneck game* if each user may route its traffic along more than a single path; it is termed an *unsplittable bottleneck game* if each user can employ only a single path. Similarly, a profile $f = \{f_p^u \,|\, u \in U, p \in P\}$ is said to be an *unsplittable flow vector* if $f_p^u \in \{0, \gamma_u\}$ for each $u \in U$ and $p \in P$ ; otherwise, *f* is said to be a *splittable flow vector*.

A profile is said to be at *Nash equilibrium* if each user considers its chosen strategy to be the best under the given choices of other users. More formally, considering a splittable (alternatively, unsplittable)

bottleneck game, $f=\left\{f_p^u\right\}$ is a *splittable (correspondingly, unsplittable) Nash flow* if, for each user $\tilde{u}\in U$ and splittable (correspondingly, unsplittable) flow vector $g=\left\{g_p^u\right\}$ that satisfies $g_p^u=f_p^u$ for each $u\in U\setminus\left\{\tilde{u}\right\}$, it holds that $b_{\tilde{u}}(f)\le b_{\tilde{u}}(g)$.

**Remark 1:** In our model, a higher bottleneck value means poorer performance. Clearly, all the results of this study can be adapted to the case where the link performance functions $\left\{q_e(f_e)\right\}$ are continuous and *decreasing* in the flows $\left\{f_e\right\}$, and the goal (of the users and the network) is to *maximize* the (corresponding) bottleneck values.

## III. EXISTENCE AND NON-UNIQUENESS OF NASH EQUILIBRIUM

In this section we establish the existence and non-uniqueness of Nash equilibria for both splittable and unsplittable bottleneck games.

### A. *Existence of Nash Equilibrium*

We begin with the splittable case.

**Theorem 1:** A splittable bottleneck game admits a Nash equilibrium.

Due to space limits, the proof appears in Appendix A.1. We note that standard techniques for proving the existence of a Nash equilibrium could not be employed for splittable bottleneck games, since the utility function of each user (i.e., $b_u(\cdot)$) is *discontinuous* (see Appendix A.8). We employ the concept of *better-reply secure* games [20], in order to overcome this difficulty.

For *unsplittable* bottleneck games, the existence of a Nash equilibrium follows directly from Theorem 3 (see Section V.A.2).

**Corollary 1:** An unsplittable bottleneck game admits a Nash equilibrium.

### B. *Non-Uniqueness of Nash Equilibria*

Through an example, we now show that, both in splittable and unsplittable bottleneck games, the Nash Equilibrium is not unique. Consider the network presented in Figure 1. Suppose that there exists a single user that needs to transfer one unit of flow from *s* to *t*, and assume that each link $e\in E$ is assigned with a performance function $q_e(f_e)=f_e$. Let $p_1=(e_1,e_3)$ and $p_2=(e_2,e_3)$. One can see that, for both splittable and unsplittable bottleneck games, the flow vectors $\left\langle f_{p_1}=1, f_{p_2}=0\right\rangle$ and $\left\langle f_{p_1}=0, f_{p_2}=1\right\rangle$ are optimal i.e., they minimize the network bottleneck. For the case of a single-user game, each of these is at Nash equilibrium.

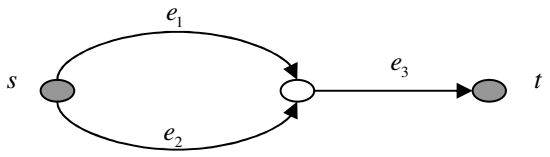



**Fig. 1: Non-uniqueness of flow at Nash Equilibrium**

## IV. CONVERGENCE TO NASH EQUILIBRIUM

In this section we address the convergence properties of *best response dynamics* in splittable and unsplittable bottleneck games. We shall focus on the *Elementary Stepwise System* (ESS) [17], in which players

update their actions sequentially, and at each update a player uses the best response action against the actions of the other players[1] . We begin with the unsplittable case.

**Theorem 2:** In unsplittable bottleneck games, starting from any flow configuration, the ESS scheme converges to a Nash equilibrium within a finite number of steps.

The proof appears in Appendix A.4.

Next, we show that, for two users, the number of steps required to reach a Nash equilibrium is polynomial. Given a pair of users $u_1, u_2$, it is easy to see that, for unsplittable flow vectors, the flow along each link can have only four different values namely, $0, \gamma_{u_1}, \gamma_{u_2}, \gamma_{u_1}+\gamma_{u_2}$ . Therefore, since (by definition) the bottleneck of each user is equal to the performance of some link in *E*, the bottlenecks of both users must be included in the set $\bigcup_{e\in E}\left\{q_e(0), q_e(\gamma_{u_1}), q_e(\gamma_{u_2}), q_e(\gamma_{u_1}+\gamma_{u_2})\right\}$, which consists of $4\cdot|E|$ elements. Finally, note that if $f_0 \to f_1 \to \ldots \to f_k$ is a sequence of converging profiles resulting from the execution of the ESS scheme, then it holds that $\max\{b_{u_1}(f_{i+2}), b_{u_2}(f_{i+2})\} < \max\{b_{u_1}(f_i), b_{u_2}(f_i)\}$ for each $i \in [0, k-2]$. Indeed, in each profile the user with the better bottleneck would never worsen the bottleneck of the other user; hence, after two profile transitions $f_i \to f_{i+1} \to f_{i+2}$, the bottleneck of the worst user must improve. Thus, since there are at most $4\cdot|E|$ possible values of the user bottleneck, the number of steps till converging to a Nash equilibrium in a two user game is at most $8\cdot|E|$.

**Remark 2:** Using similar lines, it is easy to show that the number of steps required to reach a Nash equilibrium in unsplittable bottleneck games is polynomial when the number of users is $O(1)$. Hence, in this case, The ESS scheme provides a polynomial-time algorithm for computing a Nash equilibrium.

Unfortunately, for splittable bottleneck games convergence time is unbounded even for two users, as shown by the following example.

**Example 1:** *Consider the network depicted in Fig. 2. Assume that each link $e \in E$ has a performance function $q_e(f_e) = f_e$. Assume that there are two users, A and B, each with a demand of two units. The source-destination pairs of A and B are ($s_1$,$t_1$) and ($s_2$,$t_2$), correspondingly. Suppose that the initial flow configuration is $\langle f_{p_1}=1, f_{p_2}=1, f_{p_3}=0, f_{p_4}=2\rangle$. It is easy to see that, at the $k^{th}$ step of the ESS scheme, we have $f_{p_2}^k + f_{p_3}^k = 2 - \sum_{i=0}^{k}(-1/2)^i$, for $k \in [0,\infty)$. Moreover, note that the (unique) Nash flow is $\langle f_{p_1}=4/3, f_{p_2}=2/3, f_{p_3}=2/3, f_{p_4}=4/3\rangle$; hence, a flow is at Nash equilibrium in this game only if $f_{p_2}+f_{p_3}=4/3$. However, $f_{p_2}^k + f_{p_3}^k = 2 - \sum_{i=0}^{k}(-1/2)^i \neq 4/3$ for all finite values of k.*

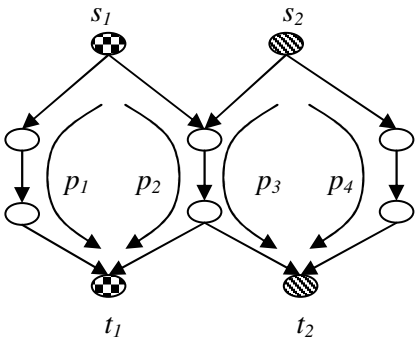



**Fig. 2: Unbounded convergence time for splittable games**

[1] We note that, if all players are allowed to move simultaneously, the system might oscillate and never reach a Nash equilibrium.

# V.(IN)EFFICIENCY OF NASH EQUILIBRIA

In this section, we investigate the degradation in network performance due to the selfish behavior of users. For both games we show that the price of anarchy is unbounded. Yet, for each game we identify conditions under which Nash equilibria are socially optimal. We begin with the unsplittable case.

## *A. (In)Efficiency of unsplittable Nash equilibria*

In the following we analyze the price of anarchy and the price of stability of unsplittable bottleneck games. While the price of anarchy is shown to be unbounded in the general case, the price of stability is shown to be 1. Thus, although a worst Nash flow may have a bottleneck value that is arbitrarily larger than the optimum, a best Nash flow always coincides with the optimum.

### *1) The price of anarchy of unsplittable bottleneck games*

We begin with a simple example that shows that the price of anarchy is unbounded for unsplittable bottleneck games.

**Example 2:** *Consider the network presented in Fig. 3, and suppose there are two users, each with the same source s and destination t. For each* $\gamma > 0$, *the first user (user A) has to transfer a demand of* $\gamma$ *units and the other user (user B) has to transfer a demand of* $2 \cdot \gamma$ *units. Finally, assume that both users must transfer their demands unsplittably. In the optimal solution, A is assigned to the upper link and B is assigned to the lower link; the corresponding bottleneck is* $\max\left\{e^{\frac{2}{3}\cdot\gamma}, e^{\frac{1}{2}\cdot 2\cdot\gamma}\right\} = e^{\gamma}$. *On the other hand, a profile where A chooses the lower link and B chooses the upper link is an unsplittable Nash flow with a bottleneck of* $\max\left\{e^{\frac{2}{3}\cdot 2\gamma}, e^{\frac{1}{2}\cdot\gamma}\right\} = e^{\frac{4\gamma}{3}}$. *Hence, the price of anarchy is* $\frac{B(f)}{B(f^*)} = \frac{e^{\frac{4\gamma}{3}}}{e^{\gamma}} = e^{\frac{\gamma}{3}}$; *as* $\gamma$ *can be arbitrarily large, this ratio is unbounded.*

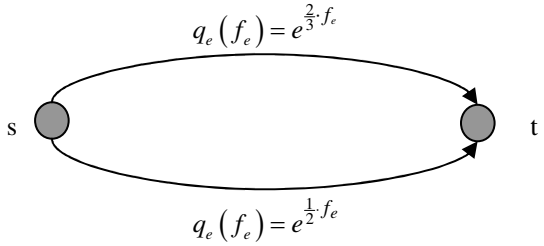



**Fig. 3: Unbounded price of anarchy for unsplittable bottleneck games**

**Remark 3:** In Appendix A.5 we show that, when all links have the same performance function $f(x) = a \cdot x^p$, where $a, p \geq 0$, then the price of anarchy is $O\left(|E|^p\right)$. We also give an example showing that this result is tight.

### *2) Stability at no price for unsplittable flows*

Although Example 2 shows that, in the worst case, unsplittable Nash equilibria can be very expensive, we now show that, in every unsplittable game, there always exists at least one Nash equilibrium that optimizes the network bottleneck. Hence, finding such an equilibrium provides an *optimal* solution that is

also *stable*, with no user having an incentive to discard it once adopted by all users. However, we prove that finding such a solution is NP-hard, even for rather simple link performance functions.

**Theorem 3:** Given a flow vector $f^*$ for the unsplittable bottleneck game $\langle G,U,\{q_e\}\rangle$, there exists a Nash flow *f* such that $B(f)\le B(f^*)$; hence, the price of stability is 1.

*Proof:* As noted in the proof of Theorem 2, in an unsplittable bottleneck game $\langle G,U,\{q_e\}\rangle$, the joint strategy space is finite i.e., the game has *finitely* many profiles (feasible unsplittable flow vectors). Given an unsplittable bottleneck game $\langle G,U,\{q_e\}\rangle$, consider its (finite) set of unsplittable flow vectors $\Theta$. For each $f\in\Theta$ consider the set of all user bottlenecks $\{b_u(f)|u\in U\}$. Let $A(f)$ be a vector that contains all the elements of $\{b_u(f)|u\in U\}$ *sorted* in a non-increasing order and let $A(f)_i$ denote the value of the *i*-th element in $A(f)$. Finally, denote by $F\subseteq\mathbb{R}^{|U|}$ the set of all vectors $A(f)$ that correspond to a feasible flow vector for the game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$ i.e., $F\triangleq\{A(f)|f$ is a feasible flow vector for the given game$\}$.

We define the relation $\prec$ over $\mathbb{R}^{|U|}$ as follows. For each $A(x),A(y)\in\mathbb{R}^{|U|}$ we say that $A(x)\prec A(y)$ *iff* there exists an $1<i\le|U|$ such that $A(x)_j=A(y)_j$ for each $j\in[1,i-1]$ and $A(x)_i<A(y)_i$. Moreover, for each $A(x)\in F$, we say that $A(x)=\min F$ if there is no $A(y)\in F$ such that $A(y)\prec A(x)$. Obviously, since we have shown that there are finitely many feasible flow vectors in the game, *F* is finite as well. Therefore, there exists $A(f)\in F$ for which there is no $A(g)\in F$ such that $A(g)\prec A(f)$; hence, there exists an $A(f)\in F$ such that $A(f)=\min F$.

Let $A(f)=\min F$. We now show that *f* is at Nash equilibrium. By definition, it holds that, for each $1<i\le|U|$, the bottleneck $A(f)_i$ of user *i* is minimized with respect to all vectors $A(g)\in F$ that satisfy $A(f)_j=A(g)_j$ for each $j\in[1,i-1]$. Thus, user *i* can improve its bottleneck *only* if the bottleneck $A(f)_j$ of at least one user $j\in[1,i-1]$ is modified. Therefore, for each $1<i\le|U|$, given the (fixed) flow of the users 1,2,…, *i*-1, user *i* cannot improve its bottleneck unless it modifies the bottleneck $A(f)_j$ of at least one user $j\in[1,i-1]$. We distinguish between two cases in which user *i* can modify the bottleneck of users in $[1,i-1]$. In the first case, user *i* increases the bottleneck $A(f)_j$ of at least one user $j\in[1,i-1]$; obviously, this can be done only if *i* ships a positive amount of traffic through one of the bottlenecks of *j*; hence, since $A(f)_j\ge A(f)_i$ for each $j\in[1,i-1]$, such a strategy can only worsen the bottleneck of *i*. In the second case, user *i* only decreases (and never increases) the bottlenecks of users in $[1,i-1]$. However, it is easy to see that such a strategy is unattainable since, by definition, it implies that there exists a feasible flow vector *g* that satisfies $A(g)\prec A(f)$; obviously, this contradicts the selection of *f* i.e., the fact that $A(f)=\min F$. Thus, we conclude that, for each $1<i\le|U|$, user *i* cannot improve its bottleneck; hence *f* is an unsplittable Nash flow.

Finally, in order to prove the theorem, we have to show that *f* produces the optimal network bottleneck among all unsplittable flows. To that end, let *g* be an unsplittable flow vector for the game $\langle G(V,E),U,\{q_e\}\rangle$. We have to show that $B(f)\le B(g)$.

Since $A(f)=\min F$, it holds by definition that the worst user bottleneck is optimized i.e., $\max_{u\in U}\{b_u(f)\}\le\max_{u\in U}\{b_u(g)\}$. Therefore, it holds that

$$\max_{e\in E|f_e>0}\{q_e(f_e)\}=\max_{u\in U}\left\{\max_{e\in E|f_e^u>0}\{q_e(f_e)\}\right\}=\max_{u\in U}\{b_u(f)\}\le\max_{u\in U}\{b_u(g)\}=\max_{u\in U}\left\{\max_{e\in E|g_e^u>0}\{q_e(g_e)\}\right\}=\max_{e\in E|g_e>0}\{q_e(g_e)\}\le\max_{e\in E}\{q_e(g_e)\}.$$

Moreover, since the performance functions are increasing, it holds that $q_e(0) \le q_e(g_e)$ for each $e \in E$; hence, $\max_{e\in E|f_e=0}\{q_e(f_e)\} = \max_{e\in E|f_e=0}\{q_e(0)\} \le \max_{e\in E}\{q_e(g_e)\}$. Thus, since both $\max_{e\in E|f_e>0}\{q_e(f_e)\} \le \max_{e\in E}\{q_e(g_e)\}$ and $\max_{e\in E|f_e=0}\{q_e(f_e)\} \le \max_{e\in E}\{q_e(g_e)\}$ hold, we conclude that $\max_{e\in E}\{q_e(f_e)\} \le \max_{e\in E}\{q_e(g_e)\}$ i.e., $B(f) \le B(g)$. ■

As explained in [1][2], the existence of "good" Nash equilibria is of major importance in the design of efficient protocols. Indeed, in many networking applications, a collective solution is proposed to all users. Therefore, it is in the interest of the protocol designer to seek the best solution that selfish users can agree upon i.e., the best Nash equilibrium. Unfortunately, while Theorem 3 guarantees that, in an unsplittable bottleneck game, there is a Nash equilibrium that is socially optimal, the following theorem shows that finding it is NP-hard.

**Theorem 4:** Given an unsplittable bottleneck game $\langle G(V,E), U, \{q_e\}\rangle$ and a value *B*, it is NP-hard to determine if the game has a Nash equilibrium with a bottleneck of at most *B*.

The proof of the theorem appears in Appendix A.6, and is based on a reduction from the *Disjoint Connecting Paths Problem* [12].

## *B. (In)Efficiency of splittable Nash equilibria*

We now turn to consider the efficiency of Nash equilibria in *splittable* bottleneck games. Again, we first quantify the price of anarchy of the game and then provide conditions under which Nash equilibria are optimal.

### *1) The price of anarchy of splittable bottleneck games*

The following example shows that the price of anarchy of splittable bottleneck games is unbounded.

**Example 3:** *Consider the network depicted in Fig. 4. Assume that each link $e \in E$ has a performance function $q_e(f_e) = 2^{f_e}$. There are two users A and B, each with a flow demand of $\gamma$ units. The source-destination pairs of A and B are ($s_1$,$t_1$) and ($s_2$,$t_2$), correspondingly. Denote $p_1 = (s_1, s_2, u, t_1)$, $p_2 = (s_1, v, t_2, t_1)$, $p_3 = (s_2, u, v, t_2)$ and $p_4 = (s_2, v, t_2)$.*

*It is easy to see that the path flow $\left\langle f_{p_1} = \frac{\gamma}{2}, f_{p_2} = \frac{\gamma}{2}, f_{p_3} = \gamma, f_{p_4} = 0 \right\rangle$ is at Nash equilibrium with a (network) bottleneck of $2^{\frac{3\cdot\gamma}{2}}$, while the path flow $\langle f_{p_1} = \gamma, f_{p_2} = 0, f_{p_3} = 0, f_{p_4} = \gamma\rangle$ is an optimal solution with a bottleneck of $2^{\gamma}$. Therefore, the ratio between the bottlenecks is $2^{\frac{\gamma}{2}}$; as $\gamma$ can be arbitrarily large, the price of anarchy is unbounded.*

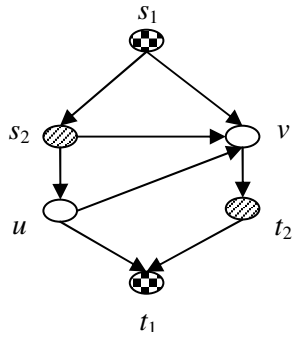



**Fig. 4: Unbounded price of anarchy for splittable bottleneck games**

### 2) *When are splittable Nash equilibria optimal?*

Note that in the (inefficient) Nash equilibrium of Example 3, user B is not routing along paths with a minimum number of bottlenecks; indeed, user B could shift its flow demand from the path $p_3$ (that contains two bottlenecks) to the path $p_4$ (that contains only one bottleneck) without affecting its performance. Thus, B unnecessarily ships traffic through an excessive number of network bottlenecks and, as a result, affects user A that shares one of these bottlenecks. On the other hand, in the optimal solution $\left\langle f_{p_1}=\gamma, f_{p_2}=0, f_{p_3}=0, f_{p_4}=\gamma \right\rangle$, users A and B are at Nash equilibrium and each routes its traffic along paths with a minimum number of bottlenecks.

In the following, we generalize the above observation to all splittable bottleneck games. More specifically, we prove that the (network) bottleneck of a splittable Nash equilibrium is optimal if the users route their traffic along paths that consist of a minimum number of bottlenecks. To that end, we need the following definition.

**Definition 1:** Given a Nash flow *f* for the splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, a pair of nodes $s,t\in V$ and a path $p\in P^{(s,t)}$, denote the number of network bottlenecks over $p$ by $N_f(p)$ i.e., $N_f(p)\triangleq\left|\left\{e\in p \middle| q_e(f)=B(f)\right\}\right|$. Then, *f* is said to *satisfy the efficiency condition* if all users route their traffic along paths with a minimum number of bottlenecks i.e., for each $u\in U$ and $p_1,p_2\in P^{(s_u,t_u)}$ with $f_{p_1}^u>0$ it holds that $N_f(p_1)\le N_f(p_2)$.

**Theorem 5:** Given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, a feasible flow vector $f^*$ and a splittable Nash flow *f* that satisfies the efficiency condition, it holds that $B(f)\le B(f^*)$; hence, a splittable Nash equilibrium that satisfies the efficiency condition is socially optimal.

Due to space limits, the proof appears in Appendix A.2.

**Remark 4:** The Braess paradox [8] shows that addition of links to a noncooperative network can negatively impact the performance of both the network and each of the users. However, for the case that $q_e(0)=0$ for each $e\in E$, it holds that the bottleneck of the network is also a bottleneck of some user in *U*. Hence, for this case, it follows from Theorem 5 that the efficiency condition is sufficient to guarantee that the paradox does not show up; indeed, at least the bottleneck of the "poorest" user (which equals the bottleneck of the network) will be minimized.

## VI. ADDITIVE NETWORK OBJECTIVES

Consider now the case in which users still have bottleneck objectives, however the network objective is *additive*, namely, it aims at minimizing the *total cost*, i.e., the sum of all link performance functions. In Appendix A.7, we show that, in this case, the price of anarchy[1] remains unbounded in both games. Nevertheless, we now show that the price of anarchy never exceeds the value of |*E*| when the users splittably route their traffic (according to bottleneck objectives) and satisfy the efficiency condition (as per Definition 1). We also show that this result is tight.

**Theorem 6:** Given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, a feasible flow vector $f^*$ and a splittable Nash flow *f* that satisfies the efficiency condition, it holds that $\sum_{e\in E} q_e(f_e)\le |E|\cdot\sum_{e\in E} q_e(f_e^*)$.

[1] I.e., the worst-case ratio between the *total cost* of a flow at Nash equilibrium and the *total cost* of an (optimal) flow that minimizes the sum of all link performance functions.

*Proof:* Since *f* satisfies the efficiency condition, it follows from Theorem 5 that the network bottleneck $B(f)=\max_{e\in E}\{q_e(f_e)\}$ is minimal (among all flow vectors that satisfy the demands of the users in *U*). In particular, $B(f)\le B(f^*)$. Therefore, by definition, $\max_{e\in E}\{q_e(f_e)\}\le\max_{e\in E}\{q_e(f_e^*)\}$. Hence, it follows that $\sum_{e\in E}q_e(f_e)\le|E|\cdot\max_{e\in E}\{q_e(f_e)\}\le|E|\cdot\max_{e\in E}\{q_e(f_e^*)\}\le|E|\cdot\sum_{e\in E}q_e(f_e^*)$. ■

We now show that this result is asymptotically tight. Specifically, we provide an example of a splittable Nash equilibrium that satisfies the efficiency condition and has a total cost that is asymptotically larger by a factor of |*E*| than the optimal cost. Consider the network presented in Fig. 5, which has |*E*|>1 parallel links. Suppose there are |*U*|>1 users, each with a single unit flow demand and the same source *s* and destination *t*. Let $0<\varepsilon<1$.

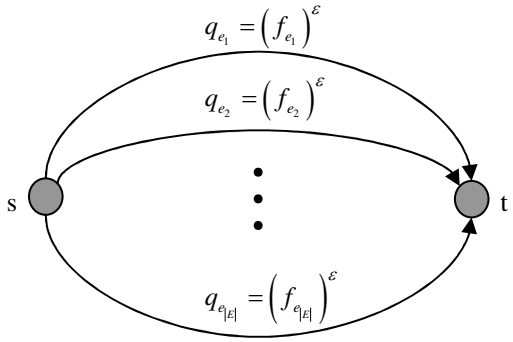



**Fig. 5: A lower bound of |*E*| for additive network objectives**

It is easy to see that a profile *f* where each user *evenly* splits its traffic between all |*E*| parallel links is a Nash flow that satisfies the efficiency conditions; the total cost of *f* is $\sum_{e\in E}q_e(f_e)=|E|\cdot\left(\frac{|U|}{|E|}\right)^{\varepsilon}$. On the other hand, in an optimal solution (i.e., one that minimizes the total cost), all users assign their traffic over the same link, resulting with a cost of $|U|^{\varepsilon}$. Hence, the ratio between the two costs is $\frac{|E|\cdot\left(\frac{|U|}{|E|}\right)^{\varepsilon}}{|U|^{\varepsilon}}\xrightarrow{\varepsilon\to 0}|E|$.

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# Appendix

## *A.1 Proof of Theorem 1*

The proof is based on a result from [20], from which we borrow some terminology and adapt it to our context (definitions A.1- A.6). Given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, denote for each user $u \in U$ the set of strategies available to *u* by $F_u$ and let $F \triangleq \underset{u\in U}{\times} F_u$. In addition, let $F_{-u}$ denote the set of strategies of all users other than *u* i.e., $F_{-u} \triangleq \underset{w\in U\setminus u}{\times} F_w$. Finally, for each user $u \in U$, let the pair $(f_u, f_{-u}) \in F_u \times F_{-u}$ denote a profile in *F* with a strategy of $f_u$ for the user *u* and a strategy profile $f_{-u}$ for all other users.

**Definition A.1:** A splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$ is *compact* if each $F_u$ is a nonempty compact subset of a topological vector space and each user bottleneck function $b_u(\cdot)$ is bounded in *F*.

**Lemma A.1***:* Splittable bottleneck games are compact.

*Proof:* Assume that we are given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$. Note that, since we assume that the source-destination nodes of all users are connected, each user has at least one strategy for routing its flow demand; hence, $F_u \neq \phi$ for each $u \in U$.

Next we show that, for each $u \in U$, the set $F_u$ is a compact subset of $\mathbb{R}^{|P|}$. To that end, note that, for each $u \in U$, $F_u$ is the space of solutions of the following system (A1)-(A3):

$$\sum_{p\in P^{(s_u,t_u)}} f_p^u = \gamma_u \qquad \text{(A1)}$$

$$f_p^u \geq 0 \qquad \forall p \in P^{(s_u,t_u)} \qquad \text{(A2)}$$

$$f_p^u = 0 \qquad \forall p \in P \setminus P^{(s_u,t_u)} \qquad \text{(A3)}$$

In order to prove that the set $F_u \subseteq \mathbb{R}^{|P|}$ is compact, it is sufficient to show that $F_u$ is bounded and closed [24]. We first show that $F_u$ is bounded. To that end, note that, for each strategy $f_u \in F_u$ in the solution space of (A1)-(A3), it holds that $0 \leq f_p^u \leq \gamma_u$ for each $p \in P$. Therefore, for each two strategies $h_u, g_u \in F_u$, it holds that $\|h_u - g_u\| \leq \sqrt{|P|}\cdot\gamma_u$. We turn to show that $F_u$ is closed. To that end, note that the solution space of (A1)-(A3) is the surface of some polyhedron in $\mathbb{R}^{|P|}$, which is closed in $\mathbb{R}^{|P|}$ [24]. Therefore, the set $F_u \subseteq \mathbb{R}^{|P|}$ is compact.

It is left to be shown that the bottleneck function $b_u(\cdot)$ is bounded on *F* for each $u \in U$. To that end, note that $b_u(f) = \max_{e\in E\left|f_e^u>0\right.} \{q_e(f)\} \leq \max_{e\in E}\{q_e(f)\}$ for each $f \in F$. Moreover, since the function $\max(\cdot)$ is continuous in the Euclidean space $\mathbb{R}^k$, and since for each $e \in E$ we assume that $q_e(\cdot)$ is continuous, it follows that the function $\max_{e\in E}\{q_e(\cdot)\}$ is continuous[1]. Therefore, since *F* is compact (as a finite Cartesian

[1] In Appendix A.8 we show that the function $b_u(f) = \max_{e\in E\left|f_e^u>0\right.} \{q_e(f_e)\}$ is, in general, discontinuous in the strategy space. However, here we consider the function $\max_{e\in E}\{q_e(f_e)\}$, in which the *max* operation is defined over the (*strategy-independent*) set *E*.

product of compact metric spaces) $\max_{e\in E}\{q_e(\cdot)\}$ *attains* the maximum over $F$ [24]; hence, there exists an $M>0$ such that $b_u(f)\le\max_{e\in E}\{q_e(f)\}<M$ for each $u\in U$. ■

**Definition A.2:** A user *u* can *secure* a bottleneck $b\in\mathbb{R}$ at a profile $f\in F$ if there exists a reply $\overline{f_u}\in F_u$, such that $b_u\left(\overline{f_u},\widetilde{f_{-u}}\right)\le b$ for all $\widetilde{f_{-u}}$ in some neighborhood of $f_{-u}$.

In other words, a bottleneck can be *secured* by a user *u* at $f\in F$ if *u* has a strategy that guarantees that bottleneck even if the other users slightly deviate from *f*.

For the next definition, we introduce the following notation. Given a profile $f\in F$, denote the vector of all user bottlenecks by $\overline{b(f)}$ i.e., $\overline{b(f)}\triangleq\left(b_{u_1}(f),b_{u_2}(f),\cdots b_{u_{|U|}}(f)\right)$. The *set of the user bottleneck vectors* is the subset of $F\times\mathbb{R}^{|U|}$ given by $\left\{\left(f,\overline{b(f)}\right)|f\in F\right\}$.

**Definition A.3:** Given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, the pair $\left(f^*,b^*\right)\in F\times\mathbb{R}^{|U|}$ is in the *closure* of the user bottleneck vector set if there exists a sequence of profiles $\left\{f^n\right\}\subseteq F$ such that $\left(f^*,b^*\right)=\lim_{n\to\infty}\left(f^n,\overline{b\left(f^n\right)}\right)$.

Considering Definition A.3, we will later show that $f^*\in F$; however, it is important to note that, since the user bottleneck function is discontinuous, it is possible that $b^*\neq\overline{b\left(f^*\right)}$.

**Definition A.4:** A splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$ is *better-reply secure* if, whenever $\left(f^*,b^*\right)$ is in the closure of the user bottleneck vector graph, and $f^*$ is not an equilibrium, some user $u\in U$ can secure a bottleneck strictly below $b_u^*$ at $f^*$.

In other words, a game is *better-reply secure* if, for every non-equilibrium profile $f^*\in F$ and sequence $\left\{f^n\right\}\subseteq F$ such that $\left(f^*,b^*\right)=\lim_{n\to\infty}\left(f^n,\overline{b\left(f^n\right)}\right)$, there exists a user $u\in U$ that can attain a bottleneck below $b_u^*$ even if the others slightly deviate from $f^*$.

**Lemma A.2:** Splittable bottleneck games are better-reply secure.

*Proof:* Given a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$, let $\left(f^*,b^*\right)\in F\times\mathbb{R}^{|U|}$ be a pair in the closure of the user bottleneck vectors set. By definition, there exists a sequence $\left\{f^n\right\}\subseteq F$ such that $\left(f^*,b^*\right)=\lim_{n\to\infty}\left(f^n,\overline{b\left(f^n\right)}\right)$. Note that, since we established in Lemma A.1 that *F* is compact, we have that $f^*\in F,$ and $b_u\left(f^*\right)$ is well-defined for all *u*.

We first show that, for each $u\in U,$ it holds that $b_u\left(f^*\right)\le b_u^*$. To that end, note that, since $f^*=\lim_{n\to\infty}f^n,$ there must exist an integer $N>0,$ such that, if in the profile $f^*$ some user $u\in U$ transfers a positive flow through some link *e* (i.e., $f_e^{*u}>0$), it must also does so through each of the profiles $\left\{f^n\right\}_{N+1}^{\infty}$ (i.e., $f_e^{n,u}>0$ for each $n>N$); hence, for each $u\in U$ and $n>N$, the set of links that are employed by *u* in $f^*$ is *contained* in the set of links that are employed by *u* in $f^n$. Therefore, $b_u^*=\lim_{n\to\infty}b_u\left(f^n\right)=\lim_{n\to\infty}\max_{e\in E\left|f_e^{n,u}>0\right.}\left\{q_e\left(f_e^n\right)\right\}\ge\lim_{n\to\infty}\max_{e\in E\left|f_e^{*u}>0\right.}\left\{q_e\left(f_e^n\right)\right\}$ for each $u\in U$. Moreover, since $q_e(\cdot)$ is con-

tinuous for each $e \in E$, it holds that $\lim_{n\to\infty} q_e\left(f_e^n\right) = q_e\left(f_e^*\right)$ for each $e \in E$. Thus, it holds that $b_u^* \geq \lim_{n\to\infty} \max_{e\in E\left|f_e^{*u}>0\right.} \left\{q_e\left(f_e^n\right)\right\} = \max_{e\in E\left|f_e^{*u}>0\right.} \left\{q_e\left(f_e^*\right)\right\} = b_u\left(f^*\right)$ for each $u \in U$.

Following the definition of *better-reply security*, assume that $f^*$ is not an equilibrium point. Hence, there exists a user $u \in U$ with a better reply $g_u$, such that $b_u\left(f^*\right) > b_u\left(g_u, f_{-u}^*\right)$. Therefore, since we have just shown that $b_u\left(f^*\right) \leq b_u^*$ for each $u \in U$, it follows that

$$b_u\left(g_u, f_{-u}^*\right) < b_u^*. \tag{A4}$$

Literally, user *u* in profile $f^*$ has a reply $g_u$ such that $b_u\left(f_{-u}^*, g_u\right) < b_u^*$. We now show that the user *u* can *secure* a bottleneck below $b_u^*$ at $f^*$. To that end, it is sufficient to show that, for the reply $g_u$, there exists some neighborhood of $f_{-u}^*$, say $N\left(f_{-u}^*\right)$, such that $b_u\left(g_u, \widetilde{f_{-u}}\right) < b_u^*$ for each $\widetilde{f_{-u}} \in N\left(f_{-u}^*\right)$.

Define $\varepsilon \triangleq \dfrac{b_u^* - b_u\left(g_u, f_{-u}^*\right)}{2} > 0$. Since $q_e(\cdot)$ is continuous for each $e \in E$, there exists a neighborhood $N\left(f_{-u}^*\right)$, such that, for each $\widetilde{f_{-u}} \in N\left(f_{-u}^*\right)$ and $e \in E$, it holds that $q_e\left(g_u, \widetilde{f_{-u}}\right) - q_e\left(g_u, f_{-u}^*\right) < \varepsilon$; hence, $b_u\left(g_u, \widetilde{f_{-u}}\right) - b_u\left(g_u, f_{-u}^*\right) = \max_{e\in E\left|g_e^u>0\right.} \left\{q_e\left(g_u, \widetilde{f_{-u}}\right)\right\} - \max_{e\in E\left|g_e^u>0\right.} \left\{q_e\left(g_u, f_{-u}^*\right)\right\} < \varepsilon$ for each $\widetilde{f_{-u}} \in N\left(f_{-u}^*\right)$. Hence, it holds that $b_u\left(g_u, \widetilde{f_{-u}}\right) < b_u\left(g_u, f_{-u}^*\right) + \varepsilon = \dfrac{b_u^* + b_u\left(g_u, f_{-u}^*\right)}{2}$ for each $\widetilde{f_{-u}} \in N\left(f_{-u}^*\right)$. Therefore, from (A4), we obtain that $b_u\left(g_u, \widetilde{f_{-u}}\right) < \dfrac{b_u^* + b_u\left(g_u, f_{-u}^*\right)}{2} < \dfrac{b_u^* + b_u^*}{2} = b_u^*$ for each $\widetilde{f_{-u}} \in N\left(f_{-u}^*\right)$. Therefore, by definition, user *u* can secure a bottleneck strictly below $b_u^*$ at $f^*$. Hence, the game is better reply secure. ■

**Definition A.5:** A function $f : F \to \mathbb{R}$ is said to be *quasi-concave* on *F* if, for a fixed $\alpha \in \mathbb{R}$, the set $\left\{x \in F \left| f(x) \geq \alpha \right.\right\}$ is convex.

**Definition A.6:** A splittable bottleneck game $\langle G(V,E), U, \{q_e(\cdot)\}\rangle$ is *quasi-concave* if, for each $u \in U$, the set $F_u$ is convex and, for every $f_{-u} \in F_{-u}$, the payoff function $b_u(\,\cdot\,, f_{-u})$ is quasi-concave on $F_u$.

**Lemma A.3:** Splittable bottleneck games are quasi-concave.

*Proof:* Given is a splittable bottleneck game $\langle G(V,E), U, \{q_e(\cdot)\}\rangle$. We first show that, for each $u \in U$, the set of feasible flow vectors $F_u$ is convex. Although the proof is quite immediate, we provide the details for completeness. For each value $\lambda \in [0,1]$ and pair of strategies (feasible flow vectors) $h_u, g_u \in F_u$, it holds that $\sum_{p\in P^{(s_u,t_u)}} \lambda \cdot g_p^u + (1-\lambda)\cdot h_p^u = \lambda \cdot \sum_{p\in P^{(s_u,t_u)}} g_p^u + (1-\lambda)\cdot \sum_{p\in P^{(s_u,t_u)}} h_p^u = \lambda\cdot\gamma_u + (1-\lambda)\cdot\gamma_u = \gamma_u$. Moreover, since $h_p^u, g_p^u \geq 0$ for each $p \in P^{(s_u,t_u)}$ it holds that $\lambda \cdot h_p^u + (1-\lambda)\cdot g_p^u \geq 0$ for each $p \in P^{(s_u,t_u)}$. Therefore, $\lambda \cdot h_u + (1-\lambda)\cdot g_u$ is also a feasible flow vector i.e., $\lambda \cdot h_u + (1-\lambda)\cdot g_u \in F_u$; Thus, by definition, $F_u$ is convex [7].

We turn to show that, for each $u \in U$ and for every $f_{-u} \in F_{-u}$, the payoff function $b_u(\,\cdot\,, f_{-u})$ is quasi-concave on $F_u$. By definition, we have to show that, for each $u \in U$ and $f_{-u} \in F_{-u}$, there exists some $\alpha \in \mathbb{R}$, such that the set $\left\{f_u \in F_u \left| b_u\left(f_u\ , f_{-u}\right) \geq \alpha \right.\right\}$ is convex; in particular, it is sufficient to show that, for each $u \in U$ and $f_{-u} \in F_{-u}$, the set $\left\{f_u \in F_u \left| b_u\left(f_u, f_{-u}\right) \geq 0 \right.\right\}$ is convex. To that end, note that, since all user

bottleneck functions are non-negative (i.e., $b_u(f) \geq 0$ for each $f \in F$), it holds that, for each $u \in U$ and $f_{-u} \in F_{-u}$, $\{f_u \in F_u \,|\, b_u(f_u, f_{-u}) \geq 0\} = F_u$. Hence, since we have shown that $F_u$ is convex for each $u \in U$, we conclude that $b_u(\cdot, f_{-u})$ is quasi-concave on $F_u$ for each $u \in U$ and $f_{-u} \in F_{-u}$. ■

We are finally ready to prove the existence of a Nash equilibrium in a splittable bottleneck game.

**Theorem 1:** A splittable bottleneck game admits a Nash equilibrium.

*Proof*: In [20] it is shown that, if a game *G* is better reply secure, quasi-concave and compact, then it possesses a Nash equilibrium. This and Lemmas A.1-A.3 prove the Theorem. ■

## *A.2 Proof of Theorem 5*

We begin by introducing the following lemma. It claims that, if *f* is a splittable Nash flow, then deleting and zeroing the flows of all users that are not shipping traffic through any network bottleneck results in a flow that is at Nash equilibrium (with respect to the new game that corresponds to the reduced set of users). Moreover, the new flow satisfies the efficiency condition and keeps the bottleneck of the network unchanged. To state the lemma, we need the following definition.

**Definition A.7:** Given is a splittable bottleneck game $\langle G(V,E), U, \{q_e(\cdot)\}\rangle$ for which $f$ is a Nash flow. Denote by $\Gamma(f)$ the set of all users that ship traffic through a network bottleneck i.e., $\Gamma(f) = \{u \in U \,|\, b_u(f) = B(f)\}$.

**Lemma A.4:** Given is a splittable bottleneck game $\langle G(V,E), U, \{q_e(\cdot)\}\rangle$ for which $f$ is a Nash flow that satisfies the efficiency condition. Consider the flow function $g: P \times \Gamma(f) \to \mathbb{R}^+ \cup \{0\}$ that satisfies $f_p^u = g_p^u$ for each $u \in \Gamma(f)$ and $p \in P$. Then, *g* is a splittable flow vector that satisfies the demands of all users in $\Gamma(f)$, such that the following four properties hold:

(i) *g* is at Nash equilibrium;
(ii) *g* satisfies the efficiency conditions;
(iii) the (network) bottleneck of *g* equals to that of *f*, i.e., $B(f) = B(g)$;
(iv) the bottleneck of each user $u \in \Gamma(f)$ is equal to that of the network i.e., $b_u(g) = B(g)$ for each $u \in \Gamma(f)$.

The proof of Lemma A.4 appears in Appendix A.3. While the details are rather tedious, the basic idea is quite straightforward, namely, the removal of users not in $\Gamma(f)$ cannot change the flow on any of the bottleneck links in the network.

**Remark:** As indicated in Appendix A.3, Lemma A.4 applies only to splittable flows, and its proof makes a crucial use of the fractional nature of such flows.

We can now prove Theorem 5.

**Theorem 5:** Given a splittable bottleneck game $\langle G(V,E), U, \{q_e(\cdot)\}\rangle$, a feasible flow vector $f^*$ and a splittable Nash flow *f* that satisfies the efficiency conditions, it holds that $B(f) \leq B(f^*)$.

*Proof*: For simplicity, we assume that the traffic demand between each source-destination pair belongs to at most one user. This assumption incurs no loss of generality for splittable flows, since all the users

that ship traffic between the same source-destination pair can be regarded as a single aggregated user with a traffic demand that is equal to the sum of demands of all the corresponding users.

Consider the Nash flow $f$ for the given splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$ and the corresponding set of users $\Gamma(f)$. Let $g:P\times\Gamma(f)\to\mathbb{R}^+\cup\{0\}$ satisfy $g_p^u=f_p^u$ for each $u\in\Gamma(f)$ and $p\in P$. From Lemma A.4 it follows that $g$ is a splittable flow vector that satisfies the demands of all users in $\Gamma(f)$, such that:

(i) $g$ is a Nash flow that satisfies the efficiency condition;

(ii) $B(f)=B(g)$ i.e., the (network) bottleneck of $g$ equals that of $f$;

(iii) $b_u(g)=B(g)$ for each $u\in\Gamma(f)$ i.e., the bottleneck of each user is equal to that of the network.

We now show that the bottleneck $B(g)$ is minimum among the (network) bottlenecks of all (splittable) flow vectors that satisfy the demands of the users in $\Gamma(f)$.

Assume, by way of contradiction, that $B(g)$ is not minimum. Hence, there exists a feasible flow vector $h$ that satisfies the demands of all users in $\Gamma(f)$ such that $B(h)<B(g)$. Denote by $\overline{E}$ the set of all network bottlenecks with respect to $g$. In addition, for each $e\in\overline{E}$, denote by $P(e)$ the set of all paths that traverse through $e$.

Since $g$ is a Nash flow, *every* path $p\in P^{(s_u,t_u)}$, where $u\in\Gamma(f)$, must traverse through at least one network bottleneck from $\overline{E}$; indeed, otherwise a user $u\in\Gamma(f)$ could have decreased its bottleneck value (which equals $B(g)$) by rerouting flow into paths whose bottlenecks are lower. Therefore, $P^{(s_u,t_u)}\subseteq\bigcup_{e\in\overline{E}}P(e)$ for each $u\in\Gamma(f)$.

Since we assume that $B(h)<B(g)$, it follows that $q_e(h_e)<q_e(g_e)$ for each $e\in\overline{E}$. Therefore, since $q_e(\cdot)$ is increasing, it holds that $h_e<g_e$ for each $e\in\overline{E}$. Therefore, for each $e\in\overline{E}$ the total traffic that traverses the paths $P(e)$ is smaller in $h$ than in $g$ i.e., $\sum_{p\in P(e)}h_p<\sum_{p\in P(e)}g_p$ for each $e\in\overline{E}$. Therefore

$$\sum_{e\in\overline{E}}\sum_{p\in P(e)}h_p<\sum_{e\in\overline{E}}\sum_{p\in P(e)}g_p\,. \tag{A5}$$

Recall that, by definition 1, given a Nash flow $f$ the number of network bottlenecks over $p$ for each $p\in P$ is denoted by $N_f(p)$. For each $p\in P$, define $N(p)\triangleq\left|p\cap\overline{E}\right|$; note that, by definition, $N(p)=N_g(p)$ for each $p\in P$. Obviously, it holds that $\sum_{e\in\overline{E}}\sum_{p\in P(e)}h_p=\sum_{p\in\bigcup_{e\in\overline{E}}P(e)}h_p\cdot N(p)$ and $\sum_{e\in\overline{E}}\sum_{p\in P(e)}g_p=\sum_{p\in\bigcup_{e\in\overline{E}}P(e)}g_p\cdot N(p)$. Therefore, from (A5) we obtain that $\sum_{p\in\bigcup_{e\in\overline{E}}P(e)}h_p\cdot N(p)<\sum_{p\in\bigcup_{e\in\overline{E}}P(e)}g_p\cdot N(p)$. Hence, it holds that

$$\sum_{p\in\bigcup_{e\in\bar{E}}P(e)}\left(g_p-h_p\right)\cdot N(p)>0$$

$$\Downarrow$$

$$\sum_{u\in\Gamma(f)}\left(\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p>0\right.}\left(g_p-h_p\right)\cdot N(p)+\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p=0\right.}\left(g_p-h_p\right)\cdot N(p)\right)>0$$

$$\Downarrow$$

$$\sum_{u\in\Gamma(f)}\left(\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p>0\right.}\left(g_p-h_p\right)\cdot N(p)-\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p=0\right.}h_p\cdot N(p)\right)>0.$$

For each $u\in\Gamma(f)$, let $N_{\max}^{(s_u,t_u)}\triangleq\max\left\{N(p)\middle|p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)},g_p>0\right\}$. Since we established (item (i) in the beginning of the proof) that the efficiency condition is satisfied for $g$, it holds by definition that $N_g(p_1)\le N_g(p_2)$ for each $u\in\Gamma(f)$ and $p_1,p_2\in P^{(s_u,t_u)}$ with $g_{p_1}^u>0$. Thus, for each $u\in\Gamma(f)$, it holds that $N_{\max}^{(s_u,t_u)}\le\min\left\{N(p)\middle|p\in\bigcup_{e\in\bar{E}}P(e)\cap P^{(s_u,t_u)},g_p=0\right\}$. Therefore

$$\sum_{u\in\Gamma(f)}\left(\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p>0\right.}\left(g_p-h_p\right)\cdot N_{\max}^{(s_u,t_u)}-\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p=0\right.}h_p\cdot N_{\max}^{(s_u,t_u)}\right)>0$$

$$\Downarrow$$

$$\sum_{u\in\Gamma(f)}N_{\max}^{(s_u,t_u)}\cdot\left(\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p>0\right.}\left(g_p-h_p\right)-\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}\left|g_p=0\right.}\left(g_p-h_p\right)\right)>0$$

$$\Downarrow$$

$$\sum_{u\in\Gamma(f)}N_{\max}^{(s_u,t_u)}\cdot\left(\sum_{p\in\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}}\left(g_p-h_p\right)\right)>0$$

Finally, since we have shown that $P^{(s_u,t_u)}\subseteq\bigcup_{e\in\bar{E}}P(e)$, it holds that $\left(\bigcup_{e\in\bar{E}}P(e)\right)\cap P^{(s_u,t_u)}=P^{(s_u,t_u)}$. Therefore, it holds that

$$\sum_{u\in\Gamma(f)}N_{\max}^{(s_u,t_u)}\cdot\left(\sum_{p\in P^{(s_u,t_u)}}\left(g_p-h_p\right)\right)>0\,. \tag{A6}$$

However, since $\sum_{p\in P^{(s_u,t_u)}}g_p=\sum_{p\in P^{(s_u,t_u)}}h_p=\gamma_u$ it holds that $\sum_{p\in P^{(s_u,t_u)}}\left(g_p-h_p\right)=0$ for each $u\in U$; hence, $\sum_{u\in\Gamma(f)}N_{\max}^{(s_u,t_u)}\cdot\left(\sum_{p\in P^{(s_u,t_u)}}\left(g_p-h_p\right)\right)=0$. Obviously, the latter contradicts (A6).

We point out that, while we have shown that $g$ achieves the minimum network bottleneck when considering only the users in $\Gamma(f)$, we still need to show that $f$ achieves the minimum network bottleneck when considering all users in $U$. Denote by $f^*$ an optimal flow, i.e., one that has the smallest bottleneck with respect to all flow vectors that satisfy the demands of the users of $U$. Also, let $\mathrm{H}$ be the set of all splittable flow vectors that satisfy the demands of the users in $\Gamma(f)$. We transform the vector $f^*$ into a

flow that belongs to the set $\mathrm{H}$ by zeroing the flow of all users in $U\setminus\Gamma(f)$ i.e., by zeroing all the flows in $\left\{f_p^{*u}\,\middle|\,p\in P, u\in U\setminus\Gamma(f)\right\}$; denote the resulting flow vector by $\widetilde{f^*}$ and note that, by construction, $f_e^*\geq\widetilde{f_e^*}$ for each $e\in E$. Since $q_e(\cdot)$ is increasing, it follows that $q_e\left(f_e^*\right)\geq q_e\left(\widetilde{f_e^*}\right)$ for each $e\in E$. Therefore, by definition,

$$B\left(f^*\right)\geq B\left(\widetilde{f^*}\right). \tag{A7}$$

However, since $\widetilde{f^*}\in\mathrm{H}$, it follows that $B\left(\widetilde{f^*}\right)\geq\min\left\{B(h)\middle|h\in\mathrm{H}\right\}$. Therefore, from (A7), it follows that $B\left(f^*\right)\geq\min\left\{B(h)\middle|h\in\mathrm{H}\right\}$. However, we have shown that $B(g)$ has the minimum bottleneck value over all flow vectors that satisfy the demands of the users in $\Gamma(f)$ i.e., $B(g)=\min\left\{B(h)\middle|h\in\mathrm{H}\right\}$. Therefore, $B\left(f^*\right)\geq B(g)$. Finally, since we established (item (ii) in the beginning of the proof) that $B(g)=B(f)$, it follows that $B\left(f^*\right)\geq B(f)$; therefore, since $f^*$ is optimal, $B\left(f^*\right)=B(f)$. Thus, the theorem is established. ■

## *A.3 Proof of Lemma A.4*

**Lemma A.4:** Given is a splittable bottleneck game $\langle G(V,E),U,\{q_e(\cdot)\}\rangle$ for which $f$ is a Nash flow that satisfies the efficiency condition. Consider the flow function $g{:}P\times\Gamma(f)\to\mathbb{R}^+\cup\{0\}$ that satisfies $f_p^u=g_p^u$ for each $u\in\Gamma(f)$ and $p\in P$. Then, *g* is a splittable flow vector that satisfies the demands of all users in $\Gamma(f)$, such that the following four properties hold:

(i) *g* is at Nash equilibrium;
(ii) *g* satisfies the efficiency conditions;
(iii) the (network) bottleneck of *g* equals to that of *f*, i.e., $B(f)=B(g)$;
(iv) the bottleneck of each user $u\in\Gamma(f)$ is equal to that of the network i.e., $b_u(g)=B(g)$ for each $u\in\Gamma(f)$.

*Proof:* First, note that, since $g_p^u=f_p^u$ for each $u\in\Gamma(f)$ and $p\in P$, it holds that g is a splittable flow vector that satisfies the flow demands of all users in $\Gamma(f)$ i.e., $\sum_{p\in P^{(s_u,t_u)}}g_p^u=\sum_{p\in P^{(s_u,t_u)}}f_p^u=\gamma_u$ for each $u\in\Gamma(f)$.

Next, we show that the bottleneck of *g* equals to that of *f*, i.e., $B(f)=B(g)$. Denote by $\widetilde{E}$ the set of all links that are the bottlenecks of the network with respect to *f* i.e., $\widetilde{E}\triangleq\left\{e\in E\middle|q_e\left(f_e\right)=B\left(f\right)\right\}$. Recall that $\Gamma(f)\subseteq U$ is the set of all users that ship traffic through one or more bottlenecks from $\widetilde{E}$ in the flow vector *f*; hence, by definition, the traffic that is carried over the links of $\widetilde{E}$ belongs only to the users in $\Gamma(f)$. Therefore, since $g_p^u=f_p^u$ for each $u\in\Gamma(f)$ and $p\in P$, it holds that $f_e=g_e$ for each $e\in\widetilde{E}$. Therefore

$$q_e(g_e)=q_e(f_e)=B(f) \text{ for each } e\in\widetilde{E}; \tag{A8}$$

Therefore, by definition of $B(\cdot)$, $B(g)\geq B(f)$. Finally, since by construction of *g*, $g_e\leq f_e$ for each $e\in E$, it follows that $q_e(g_e)\leq q_e(f_e)$ for each $e\in E$; therefore $B(g)\leq B(f)$; thus, we conclude that

$$B(g)=B(f). \tag{A9}$$

We turn to prove that, in $g$, the bottleneck of each user of $\Gamma(f)$ is equal to that of the network i.e., $b_u(g)=B(g)$ for each $u\in\Gamma(f)$. To that end, consider a user $u\in\Gamma(f)$. Since $u\in\Gamma(f)$, it follows by definition that $u$ must ship (positive) traffic through at least one network bottleneck $e\in\widetilde{E}$ in the flow vector $f$. Therefore, since $g_p^u=f_p^u$ for each $u\in\Gamma(f)$ and $p\in P$, it follows that user $u$ must also ship (positive) traffic through the link $e$ in the flow vector $g$; hence, since $e\in\widetilde{E}$ it follows according to (A8) and (A9) that $q_e(g)=B(g)$. Therefore, by definition $b_u(g)=\max_{e\in E\,|\,g_e^u>0}\{q_e(g_e)\}=B(g)$.

We turn to prove that the efficiency condition is satisfied for $g$. To that end, first note that $\widetilde{E}$ is also the set of network bottlenecks with respect to $g$; indeed, $g$ is established by zeroing the flows of all users that are not shipping traffic through any network bottleneck in $f$; thus, since it follows that the performance of the links in $\widetilde{E}$ remain unchanged in $g$ and the performance of all other links is not worsen, $\widetilde{E}$ is the set of network bottlenecks also with respect to $g$. Next, since we assume that the efficiency condition is satisfied for $f$, it holds by definition that $N_f(p_1)\le N_f(p_2)$ for each $u\in U$ and $p_1,p_2\in P^{(s_u,t_u)}$ with $f_{p_1}^u>0$. Therefore, since $g_p^u=f_p^u$ for each $u\in\Gamma(f)$ and $p\in P$, it holds that $N_g(p_1)\le N_g(p_2)$ for each $u\in\Gamma(f)$ and $p_1,p_2\in P^{(s_u,t_u)}$ with $g_{p_1}^u>0$; hence, by definition, the efficiency condition is satisfied for $g$.

We turn to show that $g$ is at Nash equilibrium. To that end, recall that the flow vector $g$ is constructed out of the given Nash flow $f$ by removing the flows of all users that are not in $\Gamma(f)$. By definition, the users that are not in $\Gamma(f)$ do not transfer (positive) flow through a network bottleneck (i.e., a link in $\widetilde{E}$). Hence, their removal can improve *only* the performance of the non-bottleneck links (i.e., the links that before the removal have already better performance then the links in $\widetilde{E}$). Thus, as performance functions are continuous, if there exists a user $u\in\Gamma(f)$ that can improve its bottleneck by moving traffic to some of the links that their condition was improved after the removal of $U\setminus\Gamma(f)$, then $u$ could also improve its bottleneck before the removal by moving traffic to these (non-bottleneck) links. In other words, if there exists a user $u\in\Gamma(f)$ that can improve its bottleneck in $g$, then it could also improve its bottleneck in $f$. Obviously, the existence of such a user contradicts the fact that $f$ is at Nash equilibrium. Therefore, $g$ is a Nash flow and the Lemma is established. ■

**Why Lemma A.4 does not hold for unsplittable flows**

For unsplittable flows, the proof of the lemma breaks at the point where we prove that $g$ is a Nash flow. Specifically, the following claim requires the "infinitely divisible" property of splittable flows: "if there exists a user $u\in\Gamma(f)$ that can improve its bottleneck by moving traffic to some of the links that their condition was improved after the removal of $U\setminus\Gamma(f)$, then $u$ could also improve its bottleneck before the removal by moving traffic to these (non-bottleneck) links.". Example 2 (Section V.A.1) clearly demonstrates it.

## *A.4 Proof of Theorem 2*

**Theorem 2:** In unsplittable bottleneck games, starting from any flow configuration, the ESS scheme converges to a Nash equilibrium within a finite number of steps.

*Proof:* Note that, in an unsplittable game, each user selects one out of at most $|P|$ possible paths; hence, each user has at most $|P|$ strategies. Therefore, since there are $|U|$ users, the joint strategy space of the game is finite i.e., the game has *finitely* many profiles (feasible unsplittable flow vectors).

Let $\langle \theta_1, \theta_2, \cdots \rangle$ be the sequence of profiles obtained throughout the execution of the ESS scheme. By definition of the scheme, for each two consecutive profiles $\theta_i, \theta_{i+1}$, it holds that exactly one user in $\theta_{i+1}$ reroutes its traffic and improves its bottleneck with respect to $\theta_i$. Since the joint strategy space is finite, it follows that, after a finite number of transitions between successive profiles in $\langle \theta_1, \theta_2, \cdots \rangle$, we must encounter some profile for the second time, and, without loss of generality, let it be $\theta_1$ and let $\theta_{n+1} = \theta_1$. Consider then the sequence of profiles $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$. Assume, by way of contradiction, that the EES scheme does not converge to a Nash equilibrium in finite time. Hence, for each profile in $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$, there exists at least one user that can improve its bottleneck.

Let $\widetilde{U} \subseteq U$ be the set of users whose bottleneck is not constant over all profiles of $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$. In addition, let $u \in \widetilde{U}$ be a user that produces the worst (i.e., largest) bottleneck among all users of $\widetilde{U}$ and over all profiles of $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$. Finally, let $\Theta = \{\theta_k\}$ be the set of all profiles in $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$ for which *u* achieves the worst bottleneck. Since $u \in \widetilde{U}$, it follows that there exists at least one profile $\theta_k \in \Theta$ such that $\theta_{k-1} \notin \Theta$; let $\theta_k$ be such a profile.

In the transition from the profile $\theta_{k-1}$ to the profile $\theta_k$, there exists exactly one user $u' \in U$ that reroutes its traffic in order to improve its bottleneck. Since $u'$ improves its bottleneck, it follows that the bottleneck of *u*' is not constant in $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$; hence, $u' \in \widetilde{U}$. Next, since the bottleneck of *u* has a smaller value in $\theta_{k-1}$ than in $\theta_k$, it follows that *u*' transfers a positive amount of flow through some bottleneck of *u* in the profile $\theta_k$. Therefore, by definition, the bottleneck of *u*' in the profile $\theta_k$ is equal to that of *u*. However, since the bottleneck of *u* in the profile $\theta_k$ is maximal with respect to all users in $\widetilde{U}$ and all profiles of $\langle \theta_1, \theta_2, \cdots \theta_n, \theta_1 \rangle$, the bottleneck of the user $u' \in U$ is not improved in the transition from $\theta_{k-1}$ to $\theta_k$. This contradicts the way by which user $u'$ was selected, hence establishing the theorem. ■

## *A.5 The Price of Anarchy of Unsplittable Bottleneck Games with Polynomial Performance Functions*

We consider the price of anarchy in unsplittable bottleneck games where $q_e(f_e) = a \cdot (f_e)^p$ for each $e \in E$, where $a, p \geq 0$. We first show that the price of anarchy is $O\left(|E|^p\right)$; then we present an example that shows that this result is tight.

**Theorem A.1:** Given an unsplittable bottleneck game $\langle G(V,E), U, \{q_e\} \rangle$, where $q_e(f_e) = a \cdot (f_e)^p$ for each $e \in E$, the price of anarchy is $O\left(|E|^p\right)$.

*Proof:* Suppose $f$ and $f^*$ are an unsplittable Nash flow and an unsplittable optimal flow, respectively. It is easy to see that, since the performance functions of all links are identical, the worst possible network bottleneck can be obtained when the traffic of all users traverses some common link i.e., when the total flow through a link is $\sum_{u \in U} \gamma_u$. In that case, the network bottleneck is $a \cdot \left( \sum_{u \in U} \gamma_u \right)^p$; hence,

$$B(f) \le a \cdot \left( \sum_{u \in U} \gamma_u \right)^p . \tag{A10}$$

On the other hand, in every (feasible) unsplittable flow vector, there exists at least one link that transfers at least $\frac{1}{|E|} \cdot \sum_{u \in U} \gamma_u$ flow units. Thus, we conclude that, the network bottleneck is at least $a \cdot \left( \frac{1}{|E|} \cdot \sum_{u \in U} \gamma_u \right)^p$; hence

$$B(f^*) \ge a \cdot \left( \frac{1}{|E|} \cdot \sum_{u \in U} \gamma_u \right)^p . \tag{A11}$$

Combing (A10) and (A11), yields the following upper bound on the price of anarchy:

$$\frac{B(f)}{B(f^*)} \le \frac{a \cdot \left( \sum_{u \in U} \gamma_u \right)^p}{a \cdot \left( \frac{1}{|E|} \cdot \sum_{u \in U} \gamma_u \right)^p} = |E|^p . \quad \blacksquare$$

We proceed to show that the result of Theorem A.1 is tight. Specifically, we provide an example where the ratio between the Nash flow and the optimal flow is $\Omega\left(|E|^p\right)$.

**Example A.1:** *Consider the network presented in Fig. A.1, and suppose that there are* $|U|$ *users, each with a demand of* $\gamma$ *flow units from source s to destination t. Assume that each link* $e \in E$ *has a performance function* $q_e(f_e) = (f_e)^p$, $p \ge 0$. *It is easy to see that, when all users transfer their traffic through the path (s,* $v_1$*,* $v_2$*,* $v_3$*,…,* $v_k$*, t), they are at a (inefficient) Nash equilibrium with a network bottleneck of* $\left(|U| \cdot \gamma\right)^p$. *On the other hand, note that an optimal solution allocates a single path from* $\Pi \triangleq \left\{ (s, v_1, t), (s, v_2, v_3, t), (s, v_4, v_5, t), \cdots, (s, v_{k-1}, v_k, t) \right\}$ *to a set of at most* $\left\lceil \frac{|U|}{|\Pi|} \right\rceil$ *users from U. Obviously, the network bottleneck in that case is at most* $\left( \left\lceil \frac{|U|}{|\Pi|} \right\rceil \cdot \gamma \right)^p$. *Hence, the price of anarchy is at least* $\frac{\left(|U| \cdot \gamma\right)^p}{\left( \left\lceil \frac{|U|}{|\Pi|} \right\rceil \cdot \gamma \right)^p} = \Omega\left(|\Pi|^p\right)$. *Finally, note that* $|\Pi| = \frac{k+1}{2}$ *and* $k = \Omega(|E|)$*; hence, the price of anarchy is* $\Omega\left(|E|^p\right)$.

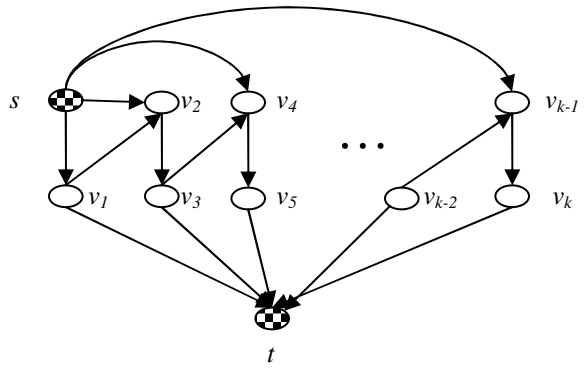



**Fig. A.1: A lower bound of $\Omega\left(|E|^p\right)$ on the price of anarchy of unsplittable bottleneck games with polynomial performance functions.**

## *A.6 Proof of Theorem 4*

**Theorem 4:** Given an unsplittable bottleneck game $\langle G(V,E),U,\{q_e\}\rangle$ and a value *B*, it is NP-hard to determine if the game has a Nash equilibrium with a bottleneck of at most *B*.

*Proof:* The reduction is from the *Disjoint Connecting Paths Problem* [12]. Given a network $G(V,E)$ and *k* distinct source-destination nodes $(s_1,t_1),(s_2,t_2),\ldots,(s_k,t_k)$, the goal of the Disjoint Connecting Paths Problem is to find *k* mutually link-disjoint paths connecting the *k* node pairs. Given an instance $\langle G(V,E),\{(s_1,t_1),(s_2,t_2),\ldots,(s_k,t_k)\}\rangle$ of the Disjoint Connecting Paths Problem, associate with each link a performance function $q_e(f_e)=f_e$ and assign to each source-destination pair a user with a demand of *B* flow units. We show that there exists an unsplittable Nash flow with a network bottleneck of at most *B iff* there exists *k* mutually disjoint paths connecting the *k* node pairs.

$\Rightarrow$: Assume first that there exists an unsplittable Nash flow with a network bottleneck of at most *B*. Then, all *k* users transfer their traffic unsplittably without intersecting with the routing paths of the other users. Hence, there exists *k* mutually disjoint paths that connect the node pairs $\{(s_1,t_1),(s_2,t_2),\ldots,(s_k,t_k)\}$.

$\Leftarrow$: Assume there exists k mutually disjoint paths connecting the k node pairs. Then, assign over each path B flow units. It is easy to see that the resulting flow is an unsplittable Nash flow with a network bottleneck of at most B. ■

## *A.7 Additive Network Objectives*

We show that, for additive network objectives, the price of anarchy[1] is unbounded in both games. Indeed, referring again to Examples 2 and 4, since users still consider bottleneck objectives, the original Nash flow remains to be such. Also, in both examples, it is easy to see that the original optimal flow (i.e., the one that minimizes the network bottleneck) also minimizes the (additive) total cost objective. Hence, the

[1] I.e., the worst case ratio between the *total cost* of a flow at Nash equilibrium and the *total cost* of optimal flow that minimizes the sum of all link performance functions.

ratio between the optimal flow (for the additive case) and the Nash flow equals $\frac{e^{\frac{4\cdot\gamma}{3}}+e^{\frac{\gamma}{2}}}{e^{\frac{2\cdot\gamma}{3}}+e^{\gamma}}$ in Example 2 and $\frac{e^{\gamma}+2\cdot e^{\frac{3\cdot\gamma}{2}}+2\cdot e^{\frac{\gamma}{2}}}{4\cdot e^{\gamma}}$ in Example 3. Since both expressions grow to infinity when $\gamma\to\infty$, the price of anarchy for additive network objectives is unbounded in both games.

## *A.8 Splittable Bottleneck Games are Discontinuous*

We now show that, since the *max* operation in the user utility function $b_u(f)=\max_{e\in E\left|f_e^u>0\right.}\left\{q_e(f_e)\right\}$ is defined over a *strategy-dependent* set, it is *discontinuous* in the user's strategy space. Indeed, for a single user that needs to transfer one unit of flow from *s* to *t* in the network of Fig. A.2, both $g=\left\langle f_{e_1}=\varepsilon, f_{e_2}=1-\varepsilon\right\rangle$ and $h=\left\langle f_{e_1}=0, f_{e_2}=1\right\rangle$ are feasible flow vectors that satisfy $\|g-h\|=\sqrt{2}\cdot\varepsilon$. However, while $\|g-h\|\to 0$ for $\varepsilon\to 0$, it holds that $b_u(g)-b_u(h)>1$ for all $\varepsilon>0$.

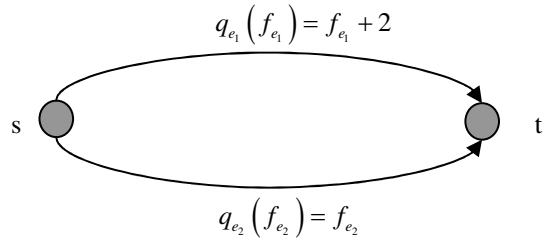



**Fig. A.2: Discontinuous user objectives in splittable bottleneck games**